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Contemporary obstacles with the old Zeno's paradoxes*

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ABSTRACT

This paper investigates the problems posed by Zeno's well-known paradoxes, or aporias regarding the notions of infinity, continuity and motion, which are still very much relevant today. First, it presents two of Zeno's paradoxes, Achilles and Arrow, and examines various proposed solutions, from Aristotle's arguments about these questions to the *Standard Solution*. It then provides an analysis of the *Continuum Hypothesis*, identified as a central challenge for the *Standard Solution*, and offers an exploration of the responses to the *Standard Solution* to highlight key objections to its underlying concepts. Finally, it offers insights into the methods that can be employed to continue to search for a solution to these paradoxes in the future.

Keywords: paradox, infinity, continuum, motion, dialectics, logic, reductio ad absurdum, regress ad infinitum.

Introduction to Zeno's paradoxes

Zeno of Elea is renowned for his method of defending the philosophical views of his teacher Parmenides, using what is known today as a logical paradox, or the

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reductio ad absurdum method. His defence of his teacher is regarding Parmenides's teachings about *the One*, that "[F8] [ii] Nor is it divisible, since it is all alike, / nor is there any more here, which would keep it from holding / together, / nor any less, but it is all full of what-is."[1] *Reductio ad absurdum* is an argumentation type that attempts to establish a thesis by deriving an absurdity from its denial, thus arguing that a proposition must be accepted because its rejection would be untenable. The result is the well-known aporias, or paradoxes, that still captivate many philosophers.

Today, we cannot know the precise formulation of Zeno's paradoxes due to the absence of any written sources of Zeno's works. We mostly know them from Aristotle and several other ancient philosophers and commentators. Still, it is not only widely accepted that his practice of articulating a clear premise that could be then used to derive logical conclusions was ground-breaking for philosophers during his time, but some of the questions that arise from the paradoxes are still of importance, especially regarding the notion of *infinity*, which modern philosophers still struggle to define.

Several of Zeno's commentators and critics have reflected on at least his two most important paradoxes, from a modern perspective, *Achilles* and *Arrow*. About these, Aristotle wrote: "This claims that the slowest runner will never be caught by the fastest runner, because the one behind has first to reach the point from which the one in front started, and so the slower one is bound always to be in front. … The third is the one I mentioned a short while ago, which claims that a moving arrow is still."[2] *Achilles* argues against the continuity of space (points), and *Arrow* against the discretion of time (moments).

The paradox of *Achilles* illustrates the problem of the divisibility of space. In this scenario, Achilles races a tortoise that he has given a headstart. When Achilles begins the race, he must first reach the halfway point of the distance the tortoise has already covered. After reaching it, he must cover half of the remaining distance, and so on indefinitely. As a result, Achilles seems stuck, needing to overcome an infinite number of halfway points between him and the tortoise before he can overtake it. The paradox lies in the fact that, in reality, Achilles can overtake the tortoise despite this infinite divisibility of space.

The *Arrow* paradox, on the other hand, addresses the nature of time. When an arrow is shot, it appears to float in the air. If one were to analyse its position in a given moment, it would appear that, in that moment, the arrow is perfectly still. According to Zeno, if an archer shoots an arrow at any moment, the arrow stays still

[1] Graham (2010), 215 – 217 // Fragment 8 // 1-52: Simplicius, *Physics*, 145. 1-146.25; 50-61; ibid. 38.30-39.9; 53-59; ibid. 30.23-31.2; 1-2: Sextus, *Against the Professors* 7 III etc. (B8).

[2] Aristoteles, *Phusike akroasis* VI, 239b 14–31 (*Physics*, English translation by Robin Waterfield).

in that moment because there cannot be movement in a single moment. However, despite the arrow being still in that moment, it is also moving towards the target. This creates a paradox: while the arrow remains still at any given moment, it ultimately reaches the target.

These paradoxes have troubled philosophers since their inception. Throughout history, many have proposed solutions, but only two approaches have been successful. The first, *Aristotle's*, dominated philosophical thought for over a millennium. The second, known as the *Standard Solution*, evolved over centuries, beginning with the work of Newton and Leibniz and evolving with Russel's contributions. In addition to these two approaches, there exists a body of objections to the *Standard Solution*, the Many objections. These objections, while not solutions in themselves, raise significant philosophical questions. Despite the consensus gentium in favour of the *Standard Solution*, they merit further exploration, especially when examined from alternative perspectives.

This paper will next examine the key philosophical questions that arise from Zeno's aporias and the three primary responses: the Aristotelian and the *Standard Solution*, and the objections to the latter. Additionally, it will analyse some of the most compelling arguments against the *Standard Solution*'s analytical approach and offer a critical assessment of the opposing perspectives. The aim is to propose an alternative framework for thinking about infinity.

Aristotle's arguments and their refutal

Zeno's paradoxes can be understood on two levels: the abstract and the practical. On an abstract level, they challenge the fundamental concepts of infinity, time, space, and motion, drawing on Parmenides' monistic teachings while opposing the Pythagorean pluralistic view of Being. Zeno highlights that these notions are poorly defined, leading to paradoxes that arise from their use. On a practical level, we have to acknowledge the human limitations in comprehending the laws of nature and reality and accept that our everyday experiences do not easily translate to a clear understanding of such abstract concepts as infinity.

At the core of Aristotle's solution to Zeno's paradoxes is his definition of infinity: "Infinity turns out to be the opposite of what people say it is."[3] He argues that infinity is not 'that which has nothing beyond itself', but rather "that which always has something beyond itself'. … According to Aristotle, something is infinite if, for

[3] Aristoteles, *Phusike akroasis* III, 206b33 (*Physics*, English translation by Robin Waterfield).

any quantity taken, further division is always possible. In contrast, something that has no further parts is considered complete and finite. Aristotle's differentiation between actual—comprising of infinitely many completed and definitive elements—and potential infinity—open for new elements, although not infinitely many—is crucial to his approach.

Aristotle concludes that the paradox of the infinite division of time and space results from assuming that either is composed of separate increments: "… in both cases, the conclusion that it is impossible to reach a limit is a result of dividing the magnitude in a certain way."[4] Regarding Arrow, in particular, the division rests on the assumption that time is composed of distinct moments of "nows": "…Here the conclusion depends on assuming that time is composed of nows; …".[5] Aristotle proposes that time is continuous, not composed of indivisible moments. Regarding Achilles, he maintains that space (in the paradox, the length of the race track) is not made up of an infinite number of points but finite segments. This very intuitive solution endured for centuries, reinforced by medieval philosophers who treated infinity as transcendent and by Aristotle's authority, especially in the fields of logic and physics.

However, the transition from Aristotle's framework of syllogistics and physics to the methodologies of Bacon and Newton marked a shift in thinking about infinity and provided a new point of view on Zeno's paradoxes. Leibniz and Newton, the pioneers of infinitesimal calculus, a subdiscipline of mathematics, introduced a new approach to continuous change that gradually replaced Aristotle's view. Leibniz, for instance, proposed that matter is divisible to infinity, arguing that every part of matter can be subdivided without end, with each part expressing motion: "... each portion of matter is not only divisible to infinity, as the ancients recognised, but also actually subdivided without end, each part into further parts, each of which one has some motion of its own: otherwise it would be impossible for each portion of matter to be able to express the whole universe."[6] This idea of infinite divisibility was a stark contrast to Aristotle's finite conception of space.

Newton's contributions further revolutionized the understanding of space and time. While Aristotle's cosmos existed within space, Newton proposed the opposite: space and time are absolute entities that contain the universe. Newton's definition of space as "absolute space, in its own nature, without relation to anything external, [that] remains always similar and immovable."[7] provided the conceptual foundation

[4] Aristoteles, *Phusike akroasis* VI, 239b18-20 (*Physics*, English translation by Robin Waterfield).

[5] Aristoteles, *Phusike akroasis* VI, 239b18-20 (*Physics*, English translation by Robin Waterfield).

[6] Strickland (2014), 27.

[7] Newton (1962), 6.

for calculus and offered a solution to Zeno's Achilles paradox. In this framework, the line between Achilles and the tortoise becomes an infinite set of points rather than a sum of parts, as Aristotle proposed. This shift paved the way for the development of the *Standard Solution* to Zeno's paradoxes.

The creation of the *Standard Solution*

The second major response to Zeno's paradoxes is what has come to be known as the *Standard Solution*. Rather than being proposed by a single philosopher, this mode of reasoning evolved gradually over centuries, drawing on the work of figures like Newton and Leibniz and, later, Dedekind, Cantor, Frege, Russel, and Whitehead. Central to this approach is the idea—consistent with modern calculus—that finite distances are indeed composed of infinitely many points and that motion, though it can be analysed in discrete moments, or what is called "at" a moment, is continuous. This mathematical framework highlights the importance of concepts from mathematics, like continuity and infinity, in addressing philosophical problems, contrary to those philosophers who argue against their relevance.

Philosophers who accepted the concept of *actual infinity* sought to address the question of how many points exist between Achilles and the tortoise. Dedekind's work on the real number line offered one way to approach this issue of continuity. He proposed that real numbers can be represented in a continuous line, segmented by what is now known as the Dedekind cut: "If the system $\mathbb{R}$ of all real numbers breaks up into two classes $\mathbb{R}_1$, $\mathbb{R}_2$ such that every number $\boldsymbol{a}_1$ of the class $\mathbb{R}_1$ is less than every number $\boldsymbol{a}_2$ of the class $\mathbb{R}_2$ then there exists one and only one number $\boldsymbol{a}$ by which this separation is produced."[8]. This idea of completeness in the real number system would allow mathematicians and philosophers to advance their understanding of infinity and continuity.

Building on Dedekind's cut, Cantor introduced the mathematical concept of transfinite sets to further explore Zeno's paradoxes: Cantor's theorem posits that every transfinite set T has parts T1 that are equivalent to it: "D: Every transfinite aggregate T is such that is has parts T1 which are equivalent to it."[9] This implies that between Achilles and the tortoise, there are infinitely many points, with a distinct point where Achilles surpasses the tortoise, leaving infinitely many points behind him. Of course, this reasoning rests on the assumption that the space described in the paradox (the

[8] Dedekind (1968), 20.

[9] Cantor (1915), 108.

race track) is Euclidean and that its physics are Newtonian—that the race path is a straight line and that the bodies moving along it can be considered material points.

However, the mathematical inquiry into the nature of infinity did not end there. The question of how many points exist on a finite line opened new problems in logic and mathematics and led to the development of Set theory, a branch of mathematics that defines sets, or collections of objects that are called members, or elements, as fundamental entities. Early Calculus alone was not sufficient to explain the infinite number of members in a set, prompting further investigations into the foundations of mathematics. Frege contributed to these efforts with his contributions to formal logic and mathematics, especially algebra. He proposed the following definition of number: "Our aim is to construct the content of a judgement which can be taken as an identity such that each side of it is a number. We are therefore proposing not to define identity specially for this case, but to use the concept of identity, taken as already known, as a means for arriving at that which is to be regarded as being identical. …"[10]

Cantor and Frege's contributions were pivotal in providing a mathematical solution to Zeno's paradoxes, particularly the continuum problem. Frege's Set theory and Cantor's Theorem are very important for Zeno's paradoxes because they define the *continuum*, or the concept of a coherent whole, that is slowly but constantly changing. However, Russell later identified a paradox in this theory that became well-known as Rusell's paradox. In his *Introduction to Mathematical Philosophy*[11], he formally expressed it as: *If* $R = \{x: x \notin x\}$, *then* $R \in R \Leftrightarrow R \notin R$. Russell resolved this problem with his Set Type Theory, which differentiates between objects, predicates, and predicates of predicates to prevent the paradox from arising. Although Russell's contribution is not universally accepted—while it clarified the nature of points on a finite line, it struggles to explain the problem of the continuity of time—it advanced the *Standard Solution*, which applies to both *Achilles* and the *Arrow*. The Arrow paradox, for instance, assumes that at each moment, the arrow is at rest because its speed at any single instant is zero. However, thanks to modern calculus and physics, we know it's not, as they allow us to calculate different speeds at each moment, resolving the paradox. In contemporary physics, time is understood not as a series of discrete instants but as continuous, enabling us to distinguish between instantaneous speed and rest. This reasoning, made possible by the development of calculus and set theory, forms the basis of the *Standard Solution* to both Achilles and the Arrow.

[10] Frege (1953), 74.

[11] Russell (1993), 136.

The Continuum Hypothesis

The many philosophers who have worked on Zeno's paradoxes disagree on many fronts, but one of the most significant is the *Continuum Hypothesis*. This hypothesis posits that there is no set whose cardinality lies between the cardinal number of integers and real numbers. In other words, any subset of the real numbers is either finite, countably infinite, or has the same cardinality as the real numbers themselves. As Cantor framed it, "The cardinal $2^{\aleph_0}$ is important since it is the size of the continuum (the set of real numbers). Cantor's famous continuum hypothesis (CH) is the statement that $2^{\aleph_0} = \aleph_1$. This is a special case of the generalized continuum hypothesis (GCH), which asserts that for all α, $2^{\aleph_\alpha} = \aleph_{\alpha+1}$."[12] Despite Cantor's efforts, the problem remains unresolved to this day, with many mathematicians and philosophers continuing to explore it.

The *Continuum Hypothesis* is crucial for the *Standard Solution* because it raises questions about the discretion or continuity of both time and space. There is a difference between the notion of *the continuum*, which can be idealized and used in mathematics, and *a continuum* that can be found in nature, which does not conform to these ideals. This distinction between the abstract continuum in mathematics and the continuum observed in nature complicates our attempts to fully grasp Zeno's paradoxes. The hypothesis also introduces the notion of looseness in the continuum, complicating our understanding of infinity even further.

The challenge of the *Continuum Hypothesis* was taken up by thinkers like Russell and Zermelo, both of whom sought to resolve it using their respective approaches. Russell, in *Principia Mathematica*, stated that "Since there is always a next higher type, we thus have a maximum cardinal in each type, without having any absolutely maximum cardinal."[13]. Zermelo extended Set Theory by adding the Axiom of Choice to the Zermelo-Fraenkel Set Theory, which states that "For every set A of pairwise-disjoint non-empty sets, there exists a set that contains exactly one element from each set in A.". Hilbert also tried to solve this problem, but he did not do it successfully.

Later contributions from Cohen and Godel further complicated matters. As they demonstrated, the *Continuum Hypothesis* (CH) "cannot be resolved on the basis of the axioms that mathematicians were employing; in modern terms, CH is independent of Zermelo-Fraenkel set theory extended with the Axiom of Choice (ZFC)."[14].

[12] Koellner (2023).

[13] Russell – Whitehead (1927), 7–8.

[14] Koellner (2023).

It can be concluded that the *Continuum Hypothesis* remains an open question, both in mathematics and philosophy.

However, unlike Zeno's, the *Continuum Hypothesis* is not a paradox that drives us into aporia. It presents a plausible explanation for the nature of the continuum, even though it cannot be definitely proven or refuted. Gödel's work, for instance, aimed to show that certain propositions, like the *Continuum Hypothesis*, cannot be axiomatically derived. As he put it: "Proposition XI: If c be a given recursive, consistent class of formulae, then the propositional formula which states that c is consistent is not c-provable; in particular, the consistency of P is unprovable in P, it being assumed that P is consistent (if not, of course, every statement is provable)."[15] Thus, while we cannot derive the *Continuum Hypothesis* from existing axioms, we can still accept it as an intuitive and plausible explanation.

The *Continuum Hypothesis* neither proves nor refutes the *Standard Solution*, but it does inform our understanding of the infinite and the continuum and shows no questions are entirely closed, whether in science or philosophy. The *Standard Solution* remains valuable for two reasons. First, it gives us the framework to analyse the notions of infinity and the continuum without relying on our own everyday experience. Second, the calculations that use this apparatus give results.

If we accept the *Standard Solution*, then we have to accept something else, too. For example, we must accept that Achilles crosses an infinite number of points in a finite time and that the continuum is too smooth to be composed of divisible points. Sometimes, accepting solutions that yield results challenges our deeply held beliefs and intuitive thoughts. Yet, it is precisely this challenge that drives the evolution of human thought and the advancement of knowledge.

The voices against the *Standard Solution*

Although there is no alternative solution that is as widely accepted as the *Standard Solution*, several philosophers have raised significant objections to it. Figures such as the dialectician Bergson, the pragmatists James and Pierce, or even the constructivist Brouwer, each from a different philosophical tradition, have tried to argue against the *Standard Solution* from their respective positions. Generally, their objections are regarding its reliance on mathematical abstraction, the limitations of mathematical representations in explaining physical reality and, in Bergson's case, the relevance of lived, intuitive experience over abstract reasoning.

[15] Gödel (1962), 70.

Bergson, a dialectician, bases his critique on his concept of *intuition*, which he defines as "the sympathy by which one is transported into the interior of an object in order to coincide with what there is unique and consequently inexpressible in it."[17] He argues that mathematical representations of space and motion, like those in the *Standard Solution*, mistakenly equate movement with the path taken. As he notes, "In the second argument (*Achilles*), movement is indeed given; it is even attributed to two moving bodies, but, always by the same error, there is an assumption that their movement coincides with their path and that we may divide it, like the path itself, in any way we please."[18] For Bergson, movement and time cannot be divided or represented in the same way as static objects, challenging the *Standard Solution*'s foundation.

Additionally, Bergson establishes the *Theory of Time*, in which time is something that cannot be statically measured because "The line one measures is immobile, time is mobility. The line is made, it is complete; time is what is happening, and more than that, it is what causes everything to happen."[18] Regarding *Arrow*, he notes: "The same fallacy appears, yet more evident, in the third argument [the *Arrow*], which consists in the conclusion that, because it is possible to distinguish points on the path of a moving body, we have the right to distinguish indivisible moments in the duration of its movement."[19] It seems that Bergson did not even come close to the discussion around the *Standard Solution*. He simply found it difficult to overcome the ever potentiality of time.

Dialecticians like Bergson mostly object to the concept of representation of space and time on a straight line with infinitely many points (moments). The first objection is towards Dedekind's cut of irrational numbers since this line is one-dimensional and space is not. The second is towards the representation of time as a one-dimensional, static, infinitely long straight line. To him, time is dynamic and cannot be measured or divided in the same way a spatial path can.

L. E. J. Brouwer, a constructivist and intuitionist philosopher, also challenges the assumptions of the *Standard Solution*, particularly its reliance on set theory and the concept of actual infinity. Brouwer's intuitionism posits that mathematical truths are not discovered but constructed by the human mind and that any mathematical object must be constructible to be meaningful. This stands in stark contrast to Cantor's transfinite set theory, which assumes the existence of infinite sets as objective truths.

[16] Bergson (1946), 189.

[17] Bergson (1991), 192.

[18] Bergson (1946), 10.

[19] Bergson (1991), 192.

Brouwer's critique is deeply rooted in his rejection of classical logic, particularly the law of the excluded middle, which allows for the acceptance of mathematical statements that cannot be constructively verified. His view leads him to reject higher-level infinite sets, such as those found in Cantor's work on cardinal numbers: "Brouwer sets out to reconstruct Cantorian set theory... [but] he realizes that this cannot be done and rejects the higher number classes, leaving only all finite ordinals and an unfinished or open-ended collection of denumerably infinite ordinals" (van Etten, 2020). By rejecting non-constructive elements of mathematics, Brouwer suggests that the *Standard Solution*, which relies on abstract notions of infinity, is insufficient for addressing Zeno's paradoxes.

Finally, the pragmatists Pierce and Robinson also expressed doubts about the *Standard Solution*. Pierce advocated for the reintroduction of infinitesimals into mathematics. While Pierce's attempt was not entirely successful because it was virtually the same as Dedekind's Theorem and because his mathematical texts about non-standard analyses were aimed at a public that was not mathematically informed, his ideas laid the groundwork for Robinson's later contributions to non-standard analysis. Both hold that philosophers should seek stronger relations between a priori knowledge and common sense and integrate intuition into tackling Zeno's paradoxes.

While these critiques do not refute the Standard Notion entirely, they highlight important limitations in its reliance on mathematical abstraction and challenge the exclusion of the intuitive and experiential and the acceptance of contra-intuitive statements about the consequences of the solution. The problem for the opposing philosophers is that we cannot overthrow intuition and experience just because the contra-intuitive solution gives results in science.

New Approaches to the old paradoxes

The problems discussed so far reveal that despite the great strides in science and the contributions to philosophical debates on infinity, continuum, and motion, it remains challenging to explain the metaphysical presuppositions about these notions. More than two millennia after Zeno's paradoxes were formulated, they remain relevant because the questions they raise challenge the core concepts in philosophy and science. This section will outline some concerns regarding the *Standard Solution* presented so far in the paper.

First, let us take a deeper look into Russell's definition of *point*: "…the least spatial thing which Geometry can deal with, the thing which, though *in* space, con-

tains no space; and this thing we define as the point."[20] Rusell acknowledges that he intentionally neglects the fundamental contradiction in this definition, that the point is metrically conceived yet *contains* no space. He later adds that the point is a result of the infinite divisibility of space "Space is continuous and infinitely divisible; the result of infinite division, the zero of extension, is called a point."[21] This introduces the problem of circular reasoning or infinite regress. For example, if space is infinitely divisible, there must exist an infinite number of points between any two different points. Our aim here is not to refute Russell's definition of *point*, but to raise awareness of the circular and infinite regress arguments that often emerge in discussions about infinity and their predisposition to become *vicious*.

However, not all regress ad infinitum arguments are inherently fallacies. As Gratton points out, "A series of steps forms a segment of an infinite regress only if (a) the steps are ordered in such a way that (b) it is possible to construct a recursive formula (c) whose output(n) (consisting of a triggering statement and $step_{(n)}$), together with that formula, entails $output_{(n+1)}$(consisting of a new triggering statement, and $step_{(n+1)}$)."[22] According to Gratton, Russell's *Axiom of Space* is not considered a vicious regress ad infinitum, as he shares later in the text: "For any natural number there is greater natural number.", i.e., $(x)(Nx \longrightarrow (\exists y)(Ny \,\&\, yGx))$, and the triggering statement, "There is the natural number 1." there follows an infinite regress that is not vicious. He does acknowledge, however, that Clark's interpretation would classify it as such."[23]

Despite the logical explanations about whether infinite regress arguments are valid or not, there remains a discomfort when applying them to "real-world" problems like motion or the nature of space. As Clark notes, "arguments that beg the question, successions of explanations that deteriorate into trivial repetitions of the same conditional justifications, each of these may involve reasoning which is logically valid. … The requirements which are flouted by these arguments are pragmatic or dialectical requirements, not formal or semantical ones."[24] In other words, even if these arguments are valid in formal logic, they fail to fully satisfy our intuitive or practical understanding of concepts like infinity or motion. This tension is one reason why the questions surrounding Zeno's paradoxes persist.

[20] Russell (1956), 128.

[21] Russell (1956), 132.

[22] Gratton (1997), 207–208.

[23] Gratton (1997), 218.

[24] Clark (1988), 370.

Even though the *Standard Solution* provides powerful answers to Zeno's aporias, some of the questions they pose remain, like those related to our intuitions about reality. The idea that a finite number of steps can resolve the problem of infinity, or that an infinite number of points can be crossed in finite time still disturbs our intuition. While these may not invalidate the *Standard Solution*, they suggest that it cannot fully account for every aspect of the debate.

It is tempting to assume that metaphysical issues such as infinity, continuum or motion can be easily discussed, but the main problems arise when we attempt to establish foundational notions and axioms that attempt to define them. This process becomes even more difficult when these axioms clash with our common-sense notions about the world. For example, one reason the *Standard Solution* is widely accepted is that it yields successful results in science and mathematics. However, as Bergson pointed out in his reflection on the Arrow Paradox, time is a dynamic process, and explaining it using static models seems inadequate.

In conclusion, when discussing the notions of infinity, continuum and motion, it's crucial to distinguish between infinity as a concept, infinite sets as mathematical structures, and infinite regress arguments. These ideas, though closely related, raise different issues that can lead the debate into vicious circles.

Metaphysical axioms or logical presuppositions

While the *Standard Solution* succeeds pragmatically, particularly in its applications to science and mathematics, it remains unsatisfactory for many philosophers because it conflicts with intuitive reasoning. This may explain why debates about motion continue to thrive both in and outside academic circles. Although this text does not delve into the specific relationship between metaphysics and logic and between philosophy and science, it is still important to consider two significant approaches to these relationships.

First, there is the question of how metaphysical presuppositions influence logical systems and second, how logic impacts metaphysical theories. The problems of the *Standard Solution* lie not so much in the capabilities of the logical apparatus or the necessity of questioning our metaphysical positions but in how they interact with notions like infinity, motion, time, space.

One way to address these paradoxes is to thread the path taken by philosophers over the past two centuries: to develop logically consistent arguments that attempt to produce solutions. Alternatively, one could adopt the perspective of those who oppose

the *Standard Solution* and question the metaphysical foundations of any proposed answer. For instance, one could ask: *Would Achilles have won the race if he overcame the Tortoise, or was he doomed to lose indefinitely? Or, similarly, Can an arrow travel if the archer does not know the target?* These questions, while not strictly logical, are extremely important for the core purpose of philosophical inquiry—to doubt the possibilities.

Some metaphysical questions may seem disconnected from logic and science, but that does not diminish their potential to enrich philosophical pluralism. Even if it's challenging to go against established theories, such metaphysical inquiries push the boundaries of what we establish as true.

Conclusion

Zeno's paradoxes remain a subject of interest because they deal with the core metaphysical concepts of infinity, continuum, and motion, which are not only abstract but also phenomena that we find in nature and struggle to understand well.

Although many philosophers have tried to provide satisfying solutions, explaining the underlying notions presents difficulties due to the complexity of their subject matter. The persistence of these problems can be seen through the very vague idea of motion, which we've shown cannot be entirely resolved using only science or mathematics.

The *Standard Solution*, while effective in practical applications, does not address all philosophical inquiries, so it is also important to ask ourselves why some philosophers find it so "unintuitive". Arsenijević would say, "The general lesson is that it is impossible *unambiguously* and *definitively* to say what path did a body cross while moving as it cannot be *unambiguous* and *definitely* said how much time has passed.[25] This reflects a deeper issue when trying to pin down fluid concepts like time, which cannot be easily fixated, and space, which cannot be easily divided: that time and space defy rigid logical frameworks.

According to Arsenijević, even in cases like Zeno's Stadium paradox, where we try to measure spatial and temporal intervals, questions like how much time has passed and how much distance was covered remain indeterminate "In the proof there is not only the relativity of movement with which all have met while driving in a car. It is about the relativity *spatial* and *temporally* intervals. It turns out that the questions "*what path have you passed while moving?*" and "*how much time had passed during that*?"

[25] Arsenijević (2007), 196.

are *not well-defined* even if the *measurement unit is fixed*."[26] Yet again, a Zeno paradox that touches on aspects of reality we encounter daily yet are reluctant to examine too closely because doing so brings us face to face with the problem—the indefiniteness of motion itself.

There is an important distinction between *consensus gentium*, or what is universally agreed upon, and *common sense*. One must always be careful when challenging *the consensus gentium* and patient to see the change, whereas it's the opposite with common sense, which requires one to have patience to challenge it and then be careful while they observe the change. This tension between consensus and intuition underpins the main methodological dispute between the proponents of the *Standard Solution* and its critics. Nevertheless, the search for new perspectives on one of Zeno's paradoxes remains valuable.

Re-examining questions that are perceived to be well-founded in science can be both constructive and enthusiastic. The act of questioning established truths requires not only enthusiasm but also the courage to pursue new lines of inquiry. While the *Standard Solution* has stood the test of time, it is by no means the final word on Zeno's paradoxes.

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[26] Arsenijević (2007), 197.

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